TITLE: INVENTORY MANAGEMENT SYSTEM

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ABSTRACT

The Inventory Management System is designed to analyze and manage raw material inventory efficiently within an organization. The system provides facilities for user registration and login for Admin and Inventory Department personnel. It helps maintain raw material master details, update stock levels, issue materials to the production department, and generate inventory-related reports.

The system allows the Admin to store and manage details of each raw material such as material name, unit, available quantity, ordering cost, inventory carrying cost, and out-of-stock cost. Based on these details, the system supports important inventory analysis techniques such as EOQ (Economic Order Quantity), FOQ (Fixed Order Quantity), and VED (Vital, Essential, Desirable) analysis. These analyses help in better inventory planning, stock control, and decision-making.

The system also supports stock issue transactions and automatically updates the available inventory. In addition, reports such as stock reports, EOQ/FOQ/VED analysis reports, delivery challan, invoice, and goods received note can be generated whenever materials are dispatched or received. The project is implemented using a relational database structure to ensure data consistency, reduced redundancy, and efficient retrieval of inventory information.

ER DIAGRAM

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The Inventory Management System consists of the following main entities:

1. USER

Stores login details of system users such as Admin and Inventory Department personnel.

Attributes:

- user_id (PK)
- username
- password
- role

2. SUPPLIER

Stores supplier details for raw materials.

Attributes:

- supplier_id (PK)
- supplier_name
- phone
- address
- email

3. RAW_MATERIAL

Stores complete details of each raw material item.

Attributes:

- material_id (PK)
- material_name
- unit
- quantity_available
- ordering_cost
- carrying_cost
- out_of_stock_cost
- supplier_id (FK)

4. STOCK_ISSUE

Stores details of materials issued to the production department.

Attributes:

- issue_id (PK)
- material_id (FK)
- issued_quantity
- issue_date
- issued_to
- issued_by

5. EOQ_ANALYSIS

Stores EOQ calculation details for each raw material.

Attributes:

- eoq_id (PK)
- material_id (FK)
- annual_demand

- ordering_cost
- carrying_cost
- eoq_value
- calculation_date

6. FOQ_ANALYSIS

Stores FOQ calculation details for each raw material.

Attributes:

- foq_id (PK)
- material_id (FK)
- fixed_order_quantity
- calculation_date

7. VED_ANALYSIS

Stores VED classification for each raw material.

Attributes:

- ved_id (PK)
- material_id (FK)
- ved_category
- remarks
- analysis_date

8. REPORT

Stores generated report details.

Attributes:

- report_id (PK)

- report_type
- generated_date
- generated_by
- material_id (FK)

9. DISPATCH

Stores material dispatch details.

Attributes:

- dispatch_id (PK)
- material_id (FK)
- quantity
- dispatch_date
- destination

10. DELIVERY_CHALLAN

Stores delivery challan details.

Attributes:

- challan_id (PK)
- dispatch_id (FK)
- challan_date
- remarks

11. INVOICE

Stores invoice details for dispatched materials.

Attributes:

- invoice_id (PK)

- dispatch_id (FK)
- invoice_date
- amount

12. GOODS_RECEIVED_NOTE

Stores GRN details for received materials.

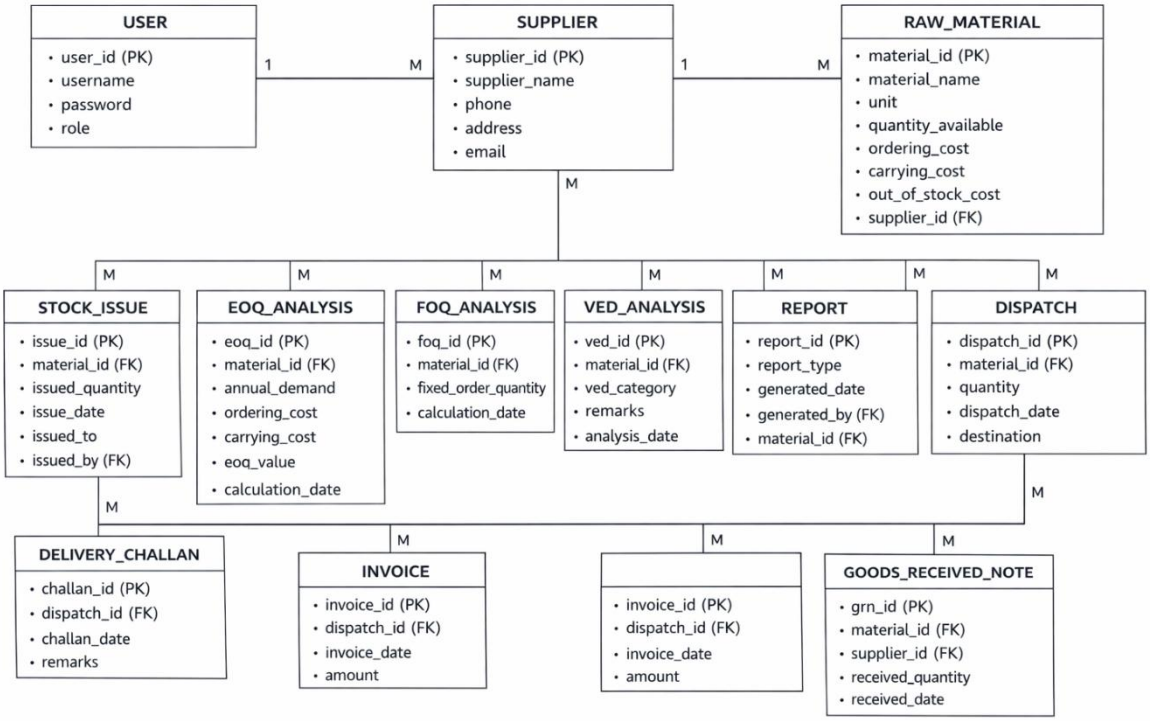
Attributes:

- grn_id (PK)
- material_id (FK)
- received_quantity
- received_date
- supplier_id (FK)

Relationships

- One supplier can supply many raw materials.
- One raw material can have many stock issue records.
- One raw material can have many EOQ analysis records.
- One raw material can have many FOQ analysis records.
- One raw material can have many VED analysis records.
- One raw material can appear in many reports.
- One dispatch record can generate one delivery challan.
- One dispatch record can generate one invoice.
- One supplier can have many goods received note records.
- One user can generate many reports.
- One user can perform many stock issue transactions.

ER Diagram of Inventory Management System



DATABASE DESIGN

The database for the Inventory Management System is designed to store user details, supplier details, raw material details, stock issue transactions, analysis records, dispatch details, and report information. The main tables used in the system are User, Supplier, Raw_Material, Stock_Issue, EOQ_Analysis, FOQ_Analysis, VED_Analysis, Report, Dispatch, Delivery_Challan, Invoice, and Goods_Received_Note.

Table 1: User Table

Field Name	Data Type	Key	Description
user_id	INT	PK	Unique ID of the user
username	VARCHAR(50)		Username for login
password	VARCHAR(100)		Password of the user
role	VARCHAR(20)		Role such as Admin or Inventory Staff

Table 2: SUPPLIER TABLE

Field Name	Data Type	Key	Description
supplier_id	INT	PK	Unique ID of supplier
supplier_name	VARCHAR(100)		Name of supplier
phone	VARCHAR(15)		Contact number
address	VARCHAR(200)		Address of supplier
email	VARCHAR(100)		Email of supplier

3. RAW_MATERIAL TABLE

Field Name	Data Type	Key	Description
material_id	INT	PK	Unique ID of raw material
material_name	VARCHAR(100)		Name of raw material
unit	VARCHAR(20)		Unit of measurement
quantity_available	INT		Quantity available in stock
ordering_cost	DOUBLE		Ordering cost of item
carrying_cost	DOUBLE		Inventory carrying cost
out_of_stock_cost	DOUBLE		Out of stock cost
supplier_id	INT	FK	Reference to supplier table

4. STOCK_ISSUE TABLE

Field Name	Data Type	Key	Description
issue_id	INT	PK	Unique ID of stock issue
material_id	INT	PK	Reference to raw material table
issued_quantity	INT		Quantity issued
issue_date	DATE		Date of issue
issued_to	VARCHAR(100)		Issued to production department
issued_by	INT	FK	Reference to user table
issued_by	INT	FK	Reference to user table

5 . EOQ_ANALYSIS TABLE

Field Name	Data Type	Key	Description
eoq_id	INT	PK	Unique ID of EOQ record
material_id	INT	FK	Reference to raw material table
annual_demand	INT		Annual demand of item
ordering_cost	DOUBLE		Ordering cost
carrying_cost	DOUBLE		Carrying cost
eoq_value	DOUBLE		Calculated EOQ value
calculation_date	DATE		Date of analysis

6. FOQ_ANALYSIS TABLE

Field Name	Data Type	Key	Description
foq_id	INT	PK	Unique ID of FOQ record
material_id	INT	FK	Reference to raw material table
fixed_order_quantity	INT		FOQ value
calculation_date	DATE		Date of analysis

7. VED_ANALYSIS TABLE

Field Name	Data Type	Key	Description
ved_id	INT	PK	Unique ID of VED record
material_id	INT	FK	Reference to raw material table
ved_category	VARCHAR(20)		Vital / Essential / Desirable
remarks	VARCHAR(100)		Remarks
analysis_date	DATE		Date of analysis

8. REPORT TABLE

Field Name	Data Type	Key	Description
report_id	INT	PK	Unique ID of report
report_type	VARCHAR(50)		Type of report
generated_date	DATE		Date of report generation
generated_by	INT	FK	Reference to user table
material_id	INT	FK	Reference to raw material table

9. DISPATCH TABLE

Field Name	Data Type	Key	Description
dispatch_id	INT	PK	Unique ID of dispatch
material_id	INT	FK	Reference to raw material table
quantity	INT		Quantity dispatched
dispatch_date	DATE		Date of dispatch
destination	VARCHAR(100)		Destination details

10. DELIVERY_CHALLAN TABLE

Field Name	Data Type	Key	Description
challan_id	INT	PK	Unique ID of challan
dispatch_id	INT	FK	Reference to dispatch table
challan_date	DATE		Date of challan generation
remarks	VARCHAR(200)		Additional remarks

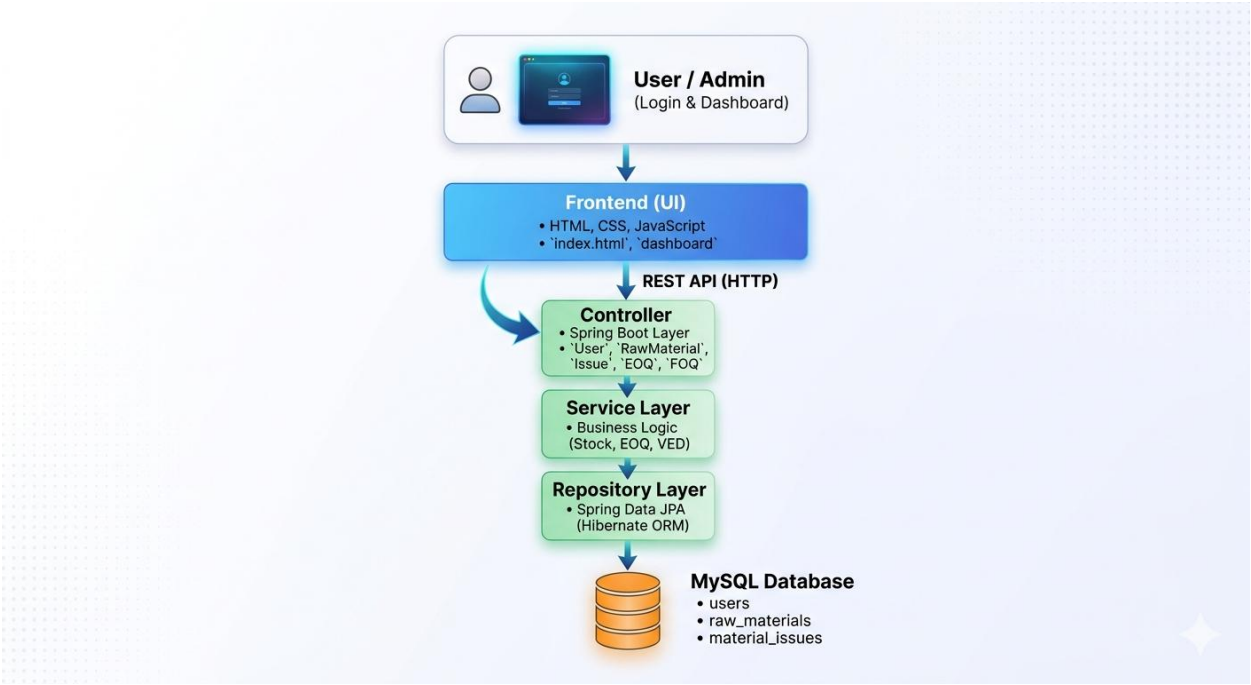
11. INVOICE TABLE

Field Name	Data Type	Key	Description
invoice_id	INT	PK	Unique ID of invoic
dispatch_id	INT	FK	Reference to dispatch table
invoice_date	DATE		Date of invoice generation
amount	DOUBLE		Total amount
amount	DOUBLE		Total amount

12. GOODS_RECEIVED_NOTE TABLE

Field Name	Data Type	Key	Description
grn_id	INT	PK	Unique ID of GRN
material_id	INT	FK	Reference to raw material table
supplier_id	INT	FK	Reference to supplier table
received_quantity	INT		Quantity received
received_date	DATE		Date of receipt

Architecture Diagram:



This architecture follows a Decoupled Layered Pattern where each section has a specific responsibility. The Frontend provides the user interface, sending requests via a REST API to the

Controller, which acts as the entry point for the backend. Inside the Service Layer, the system performs complex business calculations like EOQ and VED to manage inventory levels efficiently. The Repository Layer uses Spring Data JPA to bridge the gap between Java code and the MySQL Database, where all raw material and user data is permanently stored. This separation ensures that the application is easy to maintain, secure, and scalable as the inventory grows.